Shuyuan Zhao

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EDUCATION

Guangzhou University **Guangzhou, China**
Bachelor of Science in Statistics Sep 2023 – Jul 2027 (Expected)

- **GPA:** 3.76 / 4.00 **Rank:** 7 / 72
- **Relevant Coursework:** Probability Theory, Mathematical Statistics, Multivariate Statistical Analysis, Data Mining, Regression Analysis, Time Series Analysis.

PROFILE

Statistics undergraduate interested in applying statistical modeling, data science, and optimization-based approaches to public health and interdisciplinary decision-making. Experienced in computational research, survey research, project-based teamwork and translating real-world problems into analytical frameworks.

PUBLICATION

MANUSCRIPT UNDER REVIEW

- Lu, H., & Zhao, S. Dynamic truncation and the Berger–Boos corrected h–function method for optimal exact Poisson confidence intervals with unknown backgrounds.
- Under review at *Computational Statistics*.

RESEARCH EXPERIENCE

Dynamic truncation and the Berger−Boos corrected h−function method for optimal exact Poisson confidence intervals with unknown backgrounds

Research Member *Sep 2025 – Apr 2026*

- Investigated exact Poisson inference for rare-event data under weak signals and unknown backgrounds, with applications in high-energy physics and biomedical detection.
- Developed a dynamic truncation method using Poisson quantiles, Berger–Boos correction, and the h-function to improve coverage and interval efficiency.
- Conducted simulation studies comparing initial, one-time improved, and optimal exact confidence intervals across Bayesian, IM, and NIM methods to evaluate coverage and efficiency.
- Applied the proposed method to KARMEN neutrino detection dataset, demonstrating the proposed framework's effectiveness in practical applications.

Multidimensional Quantitative Measurement and Longitudinal Analysis of Health Risks among Middle-aged and Older Adults

Team Leader *Mar 2025 – May 2025*

- Led topic selection, research framework design, and modeling strategy, integrating Random Forest and XGBoost feature selection with fixed-effects and GEE models.
- Constructed a 6,800+ modeling sample from 21,000+ CHARLS longitudinal records across three waves, supporting machine learning, longitudinal modeling, and interpretable analysis.
- Identified 12 key frailty predictors from 26 CHARLS indicators and formed a five-dimensional frailty-risk framework covering psychological health, physical activity, economic conditions, cognitive function, and healthcare services.
- Identified age-specific shifts in dominant frailty factors from psychological to cognitive and healthcare-related risks, informing stratified intervention and community health planning.

Sustainable Tourism Optimization Model Based on the DPSIR Framework and Nonlinear Programming

Teammate, Report Writing and Modeling Analysis *Jan 2025*

- Synthesized literature and multi-source data to build a DPSIR indicator system for Juneau's overtourism challenge, covering infrastructure, ecology, residents, and external costs.
- Defined decision variables, objectives, and constraints to convert tourism sustainability trade-offs into a nonlinear optimization problem, estimating Juneau's optimal tourist capacity at 1.87 million.
- Structured the manuscript and visualized genetic-algorithm results, supporting a Juneau – Xi'an comparison and sustainable tourism recommendations.

Commercialization Plan for a Multiscale and High−Entropy Alloy Performance Prediction and Optimization Platform

Technology Commercialization and Business Plan Lead *Apr 2025 – Aug 2025*

- Translated the machine-learning-assisted high-entropy alloy platform's core technologies into a clear product proposal for aerospace and high-end manufacturing applications.
- Conducted market analysis for aerospace and high-end manufacturing applications, emphasizing the platform's commercialization value in reducing R&D cycles and experimental costs.
- Integrated technical, marketing, financial, and legal input to develop a business plan for an innovation project that later won a Provincial Silver Award.

Foshan Ancestral Temple Martial−Arts Study Tour Curriculum Design Project

Team Leader *Oct 2025 – Apr 2026*

- Led topic selection, questionnaire design, and fieldwork, using interviews, field observation, and 815 valid questionnaires responses to examine the gap between cultural recognition and public engagement.
- Modeled visitor segments with clustering and structural equation modeling, identifying differences in preferences, motivations, and dissemination intentions.
- Translated research findings into four curriculum modules, five visitor profiles, and an implementation framework integrating physical, digital, and ritual-based learning components.

HONORS & AWARDS

Honorable Mention, Mathematical Contest in Modeling (MCM/ICM) *May 2025*
National Third Prize, Zhengda Cup National Market Research and Analysis Competition *May 2025*
Provincial First Prize, National College Students Statistical Modeling Contest *Aug 2025*
Provincial Silver Award, China International College Students' Innovation Competition *Aug 2025*

ADDITIONAL EXPERIENCE

Sophia University Short-Term Program 2025 *Japan*

Participant *Jul 2025*

- Collaborated on an English group presentation in an international setting, connecting sustainable development practices with social, environmental, and urban issues.

Yiban Cloud Companionship Program

Online Volunteer Tutor *Oct 2023 – Dec 2024*

- Provided one-year weekend tutoring for a third-grade student from Shanxi, adapting mathematics explanations to his learning pace and maintaining a stable mentoring relationship.

ADDITIONAL INFORMATION

Technical Skills: R, Python, statistical modeling, machine learning, survey data analysis
Languages: Mandarin (Native), English (IELTS: 6.5)